

$$\{Q_\alpha^i, \bar{Q}_j^\beta\} = -\frac{1}{2} (\gamma_\mu)_\alpha^\beta \delta_j^i P^\mu$$

$$[M_{\mu\nu}, Q_\alpha] = -\frac{1}{2} (\gamma_{\mu\nu})_\alpha^\beta Q_\beta$$

$$[P^\mu, Q_\alpha] = 0$$

Claim: Dirac Adjoint: $\bar{\psi} := \psi^\dagger \beta = \psi^\dagger_i \gamma^0$

$$\bullet \{Q_\alpha, Q^\dagger \beta_i \gamma_0\} = -\frac{1}{2} (\gamma_\mu)_\alpha^\beta P^\mu$$

$$(Q^\dagger \gamma_0)^\beta = Q^\dagger_\delta \gamma_0^\beta_\delta$$

$$i Q_\alpha Q^\dagger_\delta (\gamma_0)^\beta_\delta + i Q^\dagger_\delta \gamma_0^\beta_\delta Q_\alpha = -\frac{1}{2} (\gamma_\mu)_\alpha^\beta P^\mu$$

$$i Q_\alpha Q^\dagger_\delta (\gamma_0)^\beta_\delta (\gamma_0)^\epsilon_\beta + i Q^\dagger_\delta \gamma_0^\beta_\delta Q_\alpha (\gamma_0)^\epsilon_\beta$$

$$= -\frac{1}{2} (\gamma_\mu)_\alpha^\beta P^\mu$$

$$i Q_\alpha (Q^\dagger)_\delta (\gamma_0^2)^\epsilon_\delta + i (Q^\dagger)_\delta (\gamma_0^2)^\epsilon_\delta Q_\alpha$$

$$= -\frac{1}{2} (\gamma_\mu)_\alpha^\beta (\gamma_0)^\epsilon_\beta P^\mu$$

$$-i Q_\alpha (Q^\dagger)^\epsilon - i (Q^\dagger)^\epsilon Q_\alpha = -\frac{1}{2} (\gamma_\mu \gamma_0)_\alpha^\epsilon P^\mu$$

$$-i \{Q_\alpha, (Q^\dagger)^\beta\} = -\frac{1}{2} (\gamma_\mu \gamma_0)_\alpha^\beta P^\mu$$

$$\{ \cdot, \cdot \}_{ce} \xrightarrow{i} \{ \cdot, \cdot \}_{Q_0}$$

$$\{Q_\alpha, (Q^\dagger)^\beta\}_{ce} = -i \frac{1}{2} (\gamma_\mu \gamma_0)_\alpha^\beta P^\mu$$

$$\rightarrow \boxed{\{Q_\alpha, (Q^\dagger)^\beta\}_{Q_0} = \frac{1}{2} (\gamma_\mu \gamma_0)_\alpha^\beta P^\mu}$$

• Taking the trace we find:

$$\text{Tr}(Q Q^\dagger + Q^\dagger Q) = \frac{1}{2} \text{Tr}(\gamma^\mu \gamma^0) P_\mu$$

where

$$\gamma^\mu \gamma^0 + \gamma^0 \gamma^\mu = 2\eta^{\mu 0} \text{Tr}(1)$$

$$\text{Tr}(1) = 4$$

$$\begin{aligned}\text{Tr}(\gamma^\mu \gamma^0) + \text{Tr}(\gamma^0 \gamma^\mu) &= 2 \text{Tr}(\gamma^\mu \gamma^0) \\ 2 \text{Tr}(\gamma^\mu \gamma^0) &= 2 \eta^{\mu 0} 4 \\ \text{Tr}(\gamma^\mu \gamma^0) &= 4 \eta^{\mu 0}\end{aligned}$$

$$\text{Tr}(Q Q^\dagger + Q^\dagger Q) = \frac{1}{2} (4 \eta^{\mu 0}) P_\mu$$

$$\therefore \boxed{\text{Tr}(Q Q^\dagger + Q^\dagger Q) = 2 P_0}$$

* Since $P^0 = E$ (Energy) we find that

$$\text{Tr}(Q Q^\dagger + Q^\dagger Q) = \sum_n \langle n | (Q Q^\dagger + Q^\dagger Q) | n \rangle = 2E$$

Since the LHS, by definition is $\langle \cdot | \cdot \rangle \geq 0$
 then so too must its sum. Hence:
 $E \geq 0$